Nithyashree Ramesh

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Summary

Pre-final year Computer Science (Cyber Security) student with experience in full-stack development, DevSecOps, and AI-driven security systems. Built scalable applications and automated vulnerability remediation pipelines using container and cloud-native technologies. Strong interest in backend systems, secure software engineering, and data structures & algorithms.

Education

BE: CSE (Cyber Security), RNS Institute of Technology **2023-2027**
CGPA: 9.1 (Current, Pre-Final Year)

Vidya Mandir Independent PU College **2021-2023**
PUC: PCMB: 93.08%

Nirmala Rani High School **2021**
SSLC: 98.08%

Technology Stack

- Languages: Python, Java, C, SQL, JavaScript
- AI / ML: LLMs, RAG, LangChain, Ollama, Agentic AI, scikit-learn, Prompt Engineering
- Cybersecurity: Container Security, SBOM, CVE Analysis, Vulnerability Triage, SaaS Security
- DevOps & Backend: Docker, Kubernetes, Flask, Node.js, PostgreSQL, MongoDB, Git, Linux
- Practices:Agile, CI/CD, Software Development Life Cycle (SDLC)

Certifications

- IBM Cybersecurity Analyst Professional Certificate - Coursera (IBM), 2024

Projects

Enhancing Product Security Using Generative AI — *Trivy, ChromaDB, RAG, KubeScore, Kubescape, Kyverno, Docker* — GitHub
- Built an end-to-end DevSecOps pipeline scanning Docker images and Kubernetes manifests using Trivy, KubeScore, Kubescape, and Kyverno, with findings stored in ChromaDB for RAG-based LLM remediation.
- Implemented iterative self-healing: each LLM-generated fix is rebuilt and rescanned until critical CVEs are resolved, reducing high/critical vulnerabilities from 15+ to ¡3 per image.
- Awarded Best Implemented Industry Project at Nokia University Day 2025, selected from 30+ competing teams.

Personalized Fitness Recommendation Platform — *MERN Stack, REST APIs*
- Developing a full-stack fitness application using the MERN stack to track workouts and nutrition.
- Implementing personalized recommendation logic for workout plans and meal suggestions based on user preferences and activity data.
- Designing RESTful APIs and backend services for user data management and progress tracking.
- Implementing JWT-based authentication, real-time updates, and analytics dashboard for user insights.

Experience

Software Testing Intern **Nov 2025 - Dec 2025**
Leistung Technologies (FireDesk NextGen) *Bengaluru, IN*
- Designed and automated 40+ end-to-end test workflows for the NextGen FireDesk MVP using Cypress, achieving 95%+ coverage across core user flows of a PERN-stack application.
- Built a real-time bug reporting pipeline to Google Sheets with automated screenshot capture on failure, enabling centralized regression tracking and eliminating manual QA updates.

Undergraduate Researcher (Institutionally Funded) **Apr 2024 - Present**
RNS Institute of Technology
- Built an ML pipeline for non-invasive migraine detection using wearable biomarkers (HRV, cortisol proxies, sleep cycles) across 3 sensor modalities.
- Trained SVM and CNN classifiers with feature engineering and cross-validation, achieving 82%+ classification accuracy on held-out data.
- Secured institutional grant funding under the Undergraduate Research Forum (URF).

Leadership & Activities

- Chair, IEEE Computer Society (2025 - Present) - Led ImpactX'25 hackathon with 35+ teams (140+ participants) and organized technical workshops impacting 100+ students.
- Joint Treasurer, IEEE Student Branch (2025 - Present) - Managed budgeting and coordinated industry speakers for large-scale events including NextGen Summit 2025.
- Volunteer, Dazzling Stars - Teaching classical music to underprivileged children (ages 6-12).